A First Course in Abstract Algebra

Section 10: Cosets and the Theorem of Lagrange

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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10 Cosets and the Theorem of Lagrange

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 10.1: Cosets of $4\mathbb{Z}$ in $\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 1 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}, \quad 3 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

Exercise 10.2: Cosets of $4\mathbb{Z}$ in $2\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $2\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

Exercise 10.3: Cosets of $\langle 2 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 2 by repeatedly adding it modulo 12:

$$H = \langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$0 + H = \{0, 2, 4, 6, 8, 10\},$$
$$1 + H = \{1, 3, 5, 7, 9, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

Exercise 10.4: Cosets of $\langle 4 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 4 by repeatedly adding it modulo 12:

$$H = \langle 4 \rangle = \{0, 4, 8\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$0 + H = \{0, 4, 8\},$$
$$1 + H = \{1, 5, 9\},$$
$$2 + H = \{2, 6, 10\},$$
$$3 + H = \{3, 7, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

Exercise 10.5: Cosets of $\langle 18 \rangle$ in $\mathbb{Z}_{36}$

First list the subgroup generated by 18 by repeatedly adding it modulo 36:

$$H = \langle 18 \rangle = \{0, 18\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$\begin{aligned}
0 + H &= \{0, 18\},\\
1 + H &= \{1, 19\},\\
2 + H &= \{2, 20\},\\
3 + H &= \{3, 21\},\\
4 + H &= \{4, 22\},\\
5 + H &= \{5, 23\},\\
6 + H &= \{6, 24\},\\
7 + H &= \{7, 25\},\\
8 + H &= \{8, 26\},\\
9 + H &= \{9, 27\},\\
10 + H &= \{10, 28\},\\
11 + H &= \{11, 29\},\\
12 + H &= \{12, 30\},\\
13 + H &= \{13, 31\},\\
14 + H &= \{14, 32\},\\
15 + H &= \{15, 33\},\\
16 + H &= \{16, 34\},\\
17 + H &= \{17, 35\}.
\end{aligned}$$

These sets are disjoint and together contain all 36 residues, verifying that the list is complete.

Exercise 10.6: Left cosets of a reflection subgroup in D_4

Let $H = \{\rho_0, \mu_2\}$. For $a \in D_4$, the left coset is $aH = \{a, a\mu_2\}$. Reading the indicated products from Table 8.12 gives

$$\begin{aligned}
\rho_0 H &= \{\rho_0, \mu_2\},\\
\rho_1 H &= \{\rho_1, \delta_2\},\\
\rho_2 H &= \{\rho_2, \mu_1\},\\
\rho_3 H &= \{\rho_3, \delta_1\}.
\end{aligned}$$

The four pairs are disjoint and include all eight group elements, so these are all the left cosets.

Exercise 10.7: Right cosets of the reflection subgroup

A right coset is $H\rho_i = \{\rho_i, \mu_2\rho_i\}$, so the list is

$$\begin{aligned} H\rho_0 &= \{\rho_0, \mu_2\}, \\ H\rho_1 &= \{\rho_1, \delta_1\}, \\ H\rho_2 &= \{\rho_2, \mu_1\}, \\ H\rho_3 &= \{\rho_3, \delta_2\}. \end{aligned}$$

These do not give the same partition as the left cosets: for example $\rho_1 H = \{\rho_1, \delta_2\}$ while $H\rho_1 = \{\rho_1, \delta_1\}$. Thus H is not normal.

Exercise 10.8: Reordered table for the reflection subgroup

Order the rows and columns in consecutive left cosets from Exercise 6. The resulting table is

	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
ρ_0	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
μ_2	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1
ρ_1	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2
δ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2
ρ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2
μ_1	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3
ρ_3	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1
δ_1	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0

The block $(\rho_0 H)(\rho_1 H)$ contains elements from different left cosets: $\rho_0\rho_1 = \rho_1 \in \rho_1 H$, but $\mu_2\rho_1 = \delta_1 \in \rho_3 H$. Both ρ_0 and μ_2 represent the same coset H, so this demonstrates that changing a representative changes the product coset. Thus coset multiplication is not well defined. There is no coset group here, so it is isomorphic to neither $\mathbb{Z}_4$ nor V.

Exercise 10.9: Left cosets of the half-turn subgroup

Set $K = \{\rho_0, \rho_2\}$. The distinct left cosets are

$$\begin{aligned} K = \{\rho_0, \rho_2\}, \quad \rho_1 K = \{\rho_1, \rho_3\}, \\ \mu_1 K = \{\mu_1, \mu_2\}, \quad \delta_1 K = \{\delta_1, \delta_2\}. \end{aligned}$$

Each pair comes from multiplying the representative by ρ_0 and by ρ_2, and the four pairs partition the eight elements.

Exercise 10.10: Right cosets of the half-turn subgroup

The half-turn ρ_2 commutes with every square symmetry: algebraically, $r^2 r^i = r^i r^2$ and $r^2(r^i s) = (r^i s)r^2$, using $sr^2 = r^{-2}s = r^2 s$. Hence $aK = Ka$ for each $a \in D_4$. The right cosets are therefore the same four pairs listed in Exercise 9; K is normal.

Exercise 10.11: Reordered table and the Klein quotient

Using the order $\rho_0, \rho_2, \rho_1, \rho_3, \mu_1, \mu_2, \delta_1, \delta_2$, the table is

	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_2	ρ_2	ρ_0	ρ_3	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_1	ρ_1	ρ_3	ρ_2	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_3	ρ_3	ρ_1	ρ_0	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	μ_2	δ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	μ_1	δ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	δ_2	μ_1	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	δ_1	μ_2	μ_1	ρ_3	ρ_1	ρ_2	ρ_0

Every 2×2 block lies entirely in a single coset, so multiplication descends to the four cosets. Label them $K, A = \rho_1 K, B = \mu_1 K, C = \delta_1 K$. The induced table is

	K	A	B	C
K	K	A	B	C
A	A	K	C	B
B	B	C	K	A
C	C	B	A	K

Every nonidentity element has order 2. Thus the quotient is the Klein four-group V, not $\mathbb{Z}_4$.

Exercise 10.12: Index of $\langle 3 \rangle$ in $\mathbb{Z}_{24}$

The multiples of 3 modulo 24 are $0, 3, 6, 9, 12, 15, 18, 21$, so the subgroup has order 8. Lagrange's counting formula gives $(\mathbb{Z}_{24} : \langle 3 \rangle) = 24/8 = 3$. Equivalently the cosets are the three residue classes modulo 3.

Exercise 10.13: Index of a reflection subgroup in S_3

A reflection μ_1 has order 2, so $\langle \mu_1 \rangle = \{e, \mu_1\}$. Since $|S_3| = 6$, its index is $6/2 = 3$.

Exercise 10.14: Index of a reflection subgroup in D_4

The element μ_2 is a reflection and has order 2. Its subgroup has two elements, whereas D_4 has eight. The number of cosets is therefore $8/2 = 4$.

Exercise 10.15: Index of the subgroup generated by σ

The factors in $\sigma = (1, 2, 5, 4)(2, 3)$ overlap, so their orders cannot be combined by a least-common-multiple rule. Follow their images from right to left:

$$1 \mapsto 2 \mapsto 3 \mapsto 5 \mapsto 4 \mapsto 1.$$

Thus $\sigma = (1, 2, 3, 5, 4)$ has order 5. Therefore $|\langle \sigma \rangle| = 5$ and $(S_5 : \langle \sigma \rangle) = 5!/5 = 24$.

Exercise 10.16: Index of the subgroup generated by μ

The cycles $(1, 2, 4, 5)$ and $(3, 6)$ are disjoint and have lengths $4, 2$. Hence μ has order $\operatorname{lcm}(4, 2) = 4$. Its cyclic subgroup has four elements, and the index in S_6 is $6!/4 = 720/4 = 180$.

Exercise 10.17: Definition of a left coset

Require that H be a *subgroup* of G, not merely a subset, and that $a \in G$. Then the left coset of H containing a is $aH = \{ah : h \in H\}$. Because $e \in H$, it indeed contains $a = ae$. For an arbitrary subset, the proposed phrase "containing a" need not be true.

Exercise 10.18: Definition of index

The definition using the number of right cosets is correct. The number of left cosets is equal to it, so either convention defines the same index $(G : H)$. For finite G this number is $|G|/|H|$; the definition itself does not require finiteness.

Exercise 10.19: True or false

(a) True: aH is defined for every subgroup $H \leq G$ and every $a \in G$.

(b) True: $|G| = (G : H)|H|$, so the number of cosets divides $|G|$.

(c) True: a group of prime order is cyclic, and cyclic groups are abelian.

(d) False: the finite subgroup $\{0\}$ of $\mathbb{Z}$ has singleton cosets $n + \{0\}$.

(e) True: $H = eH$.

(f) False: for example $\mathbb{Z} \leq \mathbb{R}$ has cosets $x + \mathbb{Z}$.

(g) True: for $n > 1$, half the permutations are even, so $[S_n : A_n] = 2$.

(h) This is a statement of taste rather than a mathematical proposition; one may agree that the theorem is nice, but there is no objectively defined truth value to prove.

(i) False: V has order 4 but no element of order 4.

(j) True: if $G = \langle g \rangle$ has order m and $d \mid m$, then $g^{m/d}$ has order d.

Exercise 10.20: Different left and right partitions in an abelian group

Impossible. For every $a \in G$ and $h \in H$, commutativity gives $ah = ha$, so $aH = Ha$. Thus the left and right coset partitions must agree for every subgroup of an abelian group.

Exercise 10.21: A single coset

Take $H = G$, for example $H = G = \mathbb{Z}_6$. Every $aH = aG = G$, so there is exactly one cell in the partition. Conversely an index-1 subgroup must be all of G, since its identity coset H is the only cell.

Exercise 10.22: Six cosets in a group of order six

Take $G = \mathbb{Z}_6$ and the trivial subgroup $H = \{0\}$. Its six cosets are the six singletons, so the partition has six cells. The formula $[G : H] = 6/1$ confirms the count.

Exercise 10.23: Twelve cosets in a group of order six

Impossible. Every coset is nonempty, so a partition of a six-element set has at most six cells. Equivalently, the index would force $|H| = 6/12 = 1/2$, which is not a possible subgroup order.

Exercise 10.24: Four cosets in a group of order six

Impossible. Lagrange's formula would give $6 = 4|H|$, so $|H| = 3/2$, not an integer. Thus the index of a subgroup of an order-6 group cannot be 4.

Exercise 10.25: Synopsis of Lagrange's theorem

The left cosets of H partition finite G, and each has $|H|$ elements by the bijection $h \mapsto ah$, so $|G| = [G : H]|H|$ and $|H|$ divides $|G|$.

Exercise 10.26: The right-coset equivalence relation

The relation is $a \sim_R b$ if and only if $ab^{-1} \in H$. It is reflexive because $aa^{-1} = e \in H$. If $ab^{-1} \in H$, inverse closure gives $(ab^{-1})^{-1} = ba^{-1} \in H$, so it is symmetric. If $ab^{-1} \in H$ and $bc^{-1} \in H$, their product is $ac^{-1} \in H$, proving transitivity. Hence it is an equivalence relation; its class containing a is Ha.

Exercise 10.27: A bijection with a right coset

Define $f : H \to Hg$ by $f(h) = hg$. Its values lie in Hg by definition. If $f(h_1) = f(h_2)$, then $h_1 g = h_2 g$; multiplying on the right by g^{-1} gives $h_1 = h_2$, proving injectivity. Every $x \in Hg$ has the form hg for some $h \in H$, so $x = f(h)$ and the map is surjective. Its inverse is $x \mapsto xg^{-1}$.

Exercise 10.28: Conjugation condition implies equal cosets

Suppose $g^{-1}hg \in H$ for all $g \in G, h \in H$. If $hg \in Hg$, write $hg = g(g^{-1}hg) \in gH$, so $Hg \subseteq gH$. For the other inclusion replace g by g^{-1} in the hypothesis: $ghg^{-1} \in H$. Then $gh = (ghg^{-1})g \in Hg$, so $gH \subseteq Hg$. The two inclusions give equality.

Exercise 10.29: Equal coset partitions imply the conjugation condition

Assume the partitions agree. The left coset gH and the right coset Hg both contain g, so they must be the same cell of that common partition. For $h \in H$, the element hg is in $Hg = gH$, so $hg = gh'$ for some $h' \in H$. Multiplying on the left by g^{-1} gives $g^{-1}hg = h' \in H$, as required.

Exercise 10.30: Do equal left cosets imply equal right cosets?

No. In S_3, take $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = a(1,2) = (1,2,3)$. Since $b = ah$ for $h \in H$, $aH = bH$. If $Ha = Hb$, the right-coset criterion would give $ba^{-1} \in H$. But

$$ba^{-1} = a(1,2)a^{-1} = (3,2) \notin H.$$

Thus $Ha \neq Hb$, giving a counterexample satisfying the assumed left-coset equality.

Exercise 10.31: Membership from equal right cosets

True. Since $e \in H$, the element $b = eb$ belongs to Hb. The assumption $Ha = Hb$ then gives $b \in Ha$. This conclusion is a membership statement and does not require commutativity or normality.

Exercise 10.32: Inverting equal cosets

True. If $aH = bH$, take the inverses of every element of both sets. Since inversion is a bijection and $H^{-1} = H$,

$$(aH)^{-1} = H^{-1}a^{-1} = Ha^{-1}, \qquad (bH)^{-1} = Hb^{-1}.$$

Equal sets have equal inverse sets, so $Ha^{-1} = Hb^{-1}$.

Exercise 10.33: Squaring representatives of equal cosets

False. Use $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = (1,2,3) = a(1,2)$ as in Exercise 30. Then $aH = bH$, but $a^2 = e$, so $a^2H = H$. Meanwhile $b^2 = (1,3,2) \notin H$, so $b^2H \neq H$. Thus squaring a representative need not preserve left-coset equality when the subgroup is not normal.

Exercise 10.34: Proper subgroups of a group of order pq

By Lagrange, the order of a subgroup divides pq. If the subgroup is proper, its order is $1, p$, or q (when $p = q$, the list reduces to $1, p$). The trivial group is cyclic. Any group of prime order is cyclic: a nonidentity element generates a subgroup whose order is a divisor greater than 1 of that prime, and so generates the whole group. Hence every proper subgroup is cyclic.

Exercise 10.35: A bijection between left and right cosets

Define F on left cosets by $F(aH) = Ha^{-1}$. It is well defined because $aH = bH$ implies $Ha^{-1} = Hb^{-1}$, as proved in Exercise 32. Conversely $Ha^{-1} = Hb^{-1}$ implies $aH = bH$ after taking inverse sets, so F is injective. Every right coset Hg equals $F(g^{-1}H)$, proving surjectivity. Thus the collections have the same cardinality even for infinite groups.

Exercise 10.36: A unique involution in an abelian group of order $2n$

Existence follows by pairing nonidentity elements with their inverses in a group of even order: at least one is its own inverse. For uniqueness, suppose distinct involutions a, b exist. Since the group

is abelian, the four elements e, a, b, ab form a subgroup: all square to e, and they are distinct, since $ab = e, a, b$ would force $a = b$, $b = e$, or $a = e$. Thus 4 divides $2n$ by Lagrange, contradicting oddness of n. Hence there is exactly one element of order 2.

Exercise 10.37: No proper nontrivial subgroups

Choose $a \neq e$. Its subgroup $\langle a \rangle$ is nontrivial, so by hypothesis it equals G and G is cyclic. It cannot be infinite cyclic, because then $\langle a^2 \rangle$ is a proper nontrivial subgroup: it contains $a^2 \neq e$ but not a. Thus G has a finite order $m > 1$. If m were composite, choose a proper divisor d with $1 < d < m$; the element $a^{m/d}$ generates a subgroup of order d, again a contradiction. Therefore m is prime.

Exercise 10.38: Multiplication of subgroup indices

Let the distinct left cosets of H in G be $a_i H$ for $1 \leq i \leq r$, and those of K in H be $b_j K$ for $1 \leq j \leq s$.

1. Cover G.

Given $g \in G$, write $g = a_i h$ for some i and $h \in H$. Then $h = b_j k$ for some $j, k \in K$, so $g \in a_i b_j K$.

2. Show the rs cosets are distinct.

If $a_i b_j K = a_{i'} b_{j'} K$, representatives differ by a factor in $K \subseteq H$, so $a_i H = a_{i'} H$ and $i = i'$. Cancelling a_i then gives $b_j K = b_{j'} K$, hence $j = j'$. Therefore these are exactly rs distinct left cosets of K in G, proving $[G : K] = [G : H][H : K]$. The argument does not require G itself to be finite.

Exercise 10.39: A subgroup of index two is normal

If $g \in H$, both gH and Hg equal H. If $g \notin H$, the two left cosets are H and gH, so $gH = G \setminus H$. The right cosets likewise give $Hg = G \setminus H$. Hence $gH = Hg$ for every g, proving normality. In fact the same complement argument works without the given finiteness assumption.

Exercise 10.40: Every element satisfies $a^n = e$

Let the order of a be d. By Lagrange applied to $\langle a \rangle$, d divides $n = |G|$, so write $n = dk$. Then $a^n = (a^d)^k = e^k = e$. The order d is finite because G is finite.

Exercise 10.41: A unique representative in $[0, 1)$

Let a coset be $a + \mathbb{Z}$. Set $x = a - \lfloor a \rfloor$. Since $\lfloor a \rfloor \leq a < \lfloor a \rfloor + 1$, we have $0 \leq x < 1$; subtracting the integer $\lfloor a \rfloor$ also shows $x \in a + \mathbb{Z}$. For uniqueness, if x, y both lie in this coset and in $[0, 1)$, their difference is an integer with $-1 < x - y < 1$. The only such integer is 0, so $x = y$.

Exercise 10.42: Sine is constant on a coset

The subgroup generated by 2π in additive $\mathbb{R}$ is $2\pi\mathbb{Z}$. If y belongs to the coset $x + 2\pi\mathbb{Z}$, then $y = x + 2\pi k$ for some integer k. By the 2π-periodicity of sine, $\sin y = \sin x$. Thus assigning $x + 2\pi\mathbb{Z} \mapsto \sin x$ is independent of the representative chosen and defines a function on the coset set.

Exercise 10.43: Double cosets

(a) Reflexivity: $a = eae$ with $e \in H, K$. Symmetry: if $a = hbk$, then $b = h^{-1}ak^{-1}$, with the inverse factors still in their subgroups. Transitivity: if $a = h_1bk_1$ and $b = h_2ck_2$, then $a = (h_1h_2)c(k_2k_1)$, with $h_1h_2 \in H$ and $k_2k_1 \in K$. Thus the relation is an equivalence relation. **(b)** Its class containing a consists exactly of all hak, with $h \in H$ and $k \in K$, namely the double coset HaK. The order of the two factors on the right in the transitivity calculation matters because the group need not be abelian.

Exercise 10.44: A point stabilizer and its transport coset

(a) Let $H = S_{c,c}$. The identity fixes c; products of maps fixing c fix c; and inverses fix c too. Hence $H \leq S_A$. **(b)** If $d \neq c$, the identity does not send c to d, so it is absent from $S_{c,d}$. Therefore this set is not a subgroup. **(c)** Choose a permutation τ sending c to d, for example the transposition (c, d). Then $S_{c,d} = \tau H$. Indeed $(\tau h)(c) = \tau(c) = d$ for $h \in H$. Conversely, if $\sigma(c) = d$, then $(\tau^{-1}\sigma)(c) = c$, so $\tau^{-1}\sigma \in H$ and $\sigma \in \tau H$. Thus it is a left coset of the stabilizer.

Exercise 10.45: The subgroups of a finite cyclic group

Let $G = \langle g \rangle$ have order n. For a divisor $d \mid n$, the element $g^{n/d}$ has order d, so $H_d = \langle g^{n/d} \rangle$ is a subgroup of order d. Conversely, every subgroup of a cyclic group is cyclic. If $H = \langle g^k \rangle$ has order d, then $d = n/\gcd(n, k)$, so $\gcd(n, k) = n/d$. The subgroup generated by g^k is exactly the subgroup generated by $g^{\gcd(n,k)}$: the exponent gcd is an integer combination of n, k, and $g^n = e$. Thus $H = \langle g^{n/d} \rangle = H_d$. This proves existence and uniqueness for each divisor, and Lagrange excludes other subgroup orders.

Exercise 10.46: The divisor sum of the Euler function

Partition the n elements of $G = \langle g \rangle$ according to their exact orders. For each divisor $d \mid n$, there is one subgroup of order d, and the elements of exact order d are precisely its generators. A cyclic order-d group has $\varphi(d)$ generators, because its generator powers h^k generate exactly when $\gcd(k, d) = 1$. Therefore counting all elements gives

$$\boxed{\sum_{d|n} \varphi(d) = n.}$$

The divisor $d = 1$ contributes the identity, since $\varphi(1) = 1$.

Exercise 10.47: A bound on roots forces cyclicity

Put $n = |G|$, and for each divisor d of n let c_d count elements of exact order d. If $c_d > 0$, choose such an element a. Its cyclic subgroup has d elements, all satisfying $x^d = e$. By the hypothesis there are at most d solutions in all of G, so these are all the solutions. Thus every element of exact order d belongs to $\langle a \rangle$, and $c_d = \varphi(d)$. If no such element exists, $c_d = 0$. In either case $c_d \leq \varphi(d)$. Lagrange says every element order divides n, so

$$n = \sum_{d|n} c_d \leq \sum_{d|n} \varphi(d) = n.$$

Equality of the two sums of nonnegative terms forces equality in every term: otherwise at least one strictly smaller term and no larger terms would make the sum smaller. In particular $c_n = \varphi(n) > 0$. An element of order n generates all n elements of G, so G is cyclic.